

Tulsi Ram Pokhrel

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PROFESSIONAL SUMMARY

Graduate Research Assistant in Agricultural and Biosystems Engineering specializing in **hydrologic, hydraulic, and water-quality** modeling with over **2 years** of applied research experience and **3 years** of teaching in water-resources engineering subjects. Skilled in applying GIS, remote sensing, and data analytics for watershed management, flood assessment, and **economic analysis** of conservation practices to support data-driven policy development and sustainable land-management decisions. Adept at communicating complex technical findings to diverse audiences and delivering results in collaborative, multidisciplinary environments.

EDUCATION

Doctor of Philosophy in Agricultural and Biosystems Engineering June 2026- Present
South Dakota State University, Brookings, SD

M.S. in Agricultural and Biosystems Engineering | GPA: 4.00/4.00 May 2026
South Dakota State University, Brookings, SD

B.S. in Agricultural Engineering | GPA: 3.88/4.00 Aug 2021
Tribhuvan University, Nepal

TECHNICAL SKILLS

- **Hydrologic & Hydraulic Modeling:** SWAT, SWAT+, HEC-RAS, HEC-HMS, HY-8
- **Design & Engineering Tools:** AutoCAD, Civil 3D, Microsoft Excel (advanced modeling & VBA), Power BI (data visualization)
- **Geospatial & Remote Sensing Tools:** QGIS, ArcGIS Pro, ArcGIS Online, ENVI, Google Earth Engine, Imagine
- **Data Science & Machine Learning:** Python, R, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Keras
- **Productivity & Collaboration:** Microsoft Office (Word, Excel, PowerPoint, Outlook), Teams

PROFESSIONAL EXPERIENCE

Graduate Research Assistant (M.S.) Aug 2024 – Present
South Dakota State University, Brookings, SD

- Applied SWAT+ and HEC-RAS models to analyze watershed hydrology, flood behavior, and water-quality dynamics, supporting decision-making for sustainable land and water management
- Processed geospatial datasets (remote sensing, climate, and soil data) using ArcGIS, Google Earth Engine, and Python, and applied machine learning for predictive modeling in water resources planning
- Performed economic analysis of conservation practices to evaluate cost-effective strategies for water-quality improvement and resource sustainability.
- Developed expertise in using HYPROP, KSAT, WP4C, and PARIO for measuring soil hydraulic and physical properties to evaluate soil health under diverse land-use conditions.
- Collaborated with multidisciplinary teams and communicated research progress and findings to stakeholders and funding agencies through reports, presentations, and professional conferences

Assistant Lecturer/Lecturer (Full time) Nov 2021 – Jul 2024
Tribhuvan University, Nepal

Delivered lectures on Hydraulics, Hydrology, Fluid Mechanics, Wastewater Engineering, and Irrigation and Drainage Engineering for undergraduate civil engineering students
Designed and conducted lab sessions and field-based projects linking theoretical concepts with real-world water resources applications

- Mentored students through hands-on training in HEC-RAS and GIS-based flood modeling, assisting them in map preparation, analysis, and report writing for final-year design projects.

CONFERENCE PRESENTATIONS

POSTER PRESENTATIONS

- **Pokhrel, T. R.,** Mehan, S., Trooien, T., Gilbertson, J., and Berg, B. (2026). *Using Watershed-Scale SWAT+ Modeling and Critical Source Area Analysis to Inform Seasonal Riparian Area Management*. 2026 Western South Dakota Environmental and Hydrology Conference, Rapid City, South Dakota, United States. (2026, April 16)
- **Pokhrel, T. R.,** Mehan, S., Trooien, T., Gilbertson, J., and Berg, B. (2026). *Using Watershed-Scale SWAT+ Modeling and Critical Source Area Analysis to Inform Seasonal Riparian Area Management*. 2026 ASABE North Central Intersectional Conference, St. Paul, Minnesota, USA. (2026, April 10)
- **Pokhrel, T. R.,** Mehan, S., Trooien, T., Gilbertson, J., & Berg, B. (2025). *Informing Riparian Conservation through Watershed-Scale SWAT+ Simulation of Nitrate and Sediment*. 2026 South Dakota ASABE Section Meeting, Brookings, SD, USA (April 2, 2026) and AGU 2025 Annual Meeting, New Orleans, LA, USA (December 14-18, 2025).
- **Pokhrel, T.,** Mehan, S., Trooien, T., Gilbertson, J., & Berg, B. (2025). *Ecological and Economic Benefits of a Nature-Based Conservation Practice in Big Sioux Watershed, South Dakota*. 2025 Nebraska Water Conference, Omaha, NE, USA. (September 18-19, 2025)
- **Pokhrel, T.,** Mehan, S., Trooien, T., Gilbertson, J., & Berg, B. (2025). *Assessing the Impact of Seasonal Riparian Area Management on Vegetation Health: A Remote Sensing Trend Analysis in the Skunk Creek Riparian Zone*. UCOWR/NIWR Annual Water Resources Conference 2025, University of Minnesota, Minneapolis, MN, USA. (June 3-5, 2025)
- **Pokhrel, T.,** & Mehan, S. (2025) *Assessing the Combined Influence of Seasonal Riparian Area Management on Riparian Vegetation and Water Quality Dynamics*. 2025 Western South Dakota Hydrology Conference (April 10, 2025) and 2025 South Dakota ASABE Section Meeting, Brookings, SD, USA. (April 3, 2025)
- **Pokhrel, T.,** Mehan, S., Niroula, A., & Alam, J. (2024). *Spatial Variation of Water Balance Components of Tamor River Basin using SWAT*. 2024 South Dakota Student Water Conference, Brookings, SD, USA. (October 15, 2024)

ORAL PRESENTATIONS

- **Pokhrel, T.,** Mehan, S., Trooien, T., Gilbertson, J., & Berg, B. (2026). *Evaluating Seasonal Riparian Grazing Strategies for Sediment and Nitrate Load Reduction Using SWAT+*. ASABE Annual International Meeting 2026, Indianapolis, Indiana. (July 12-15, 2026)
- **Pokhrel, T.,** Mehan, S., Trooien, T., Gilbertson, J., & Berg, B. (2025). *Environmental and economic efficacy of the Seasonal Riparian Area Management (SRAM) program in the Big Sioux River Watershed*. Missouri River Science and Management Symposium 2025, University of South Dakota, Vermillion, SD, USA. (November 13, 2025)
- **Pokhrel, T.,** Mehan, S., Trooien, T., Gilbertson, J., & Berg, B. (2025). *Identification of Priority Zones for Riparian Buffer Enrollment under the Seasonal Riparian Area Management Program using SWAT+*. International SWAT Conference & Workshops 2025, Colorado State University, Fort Collins, Colorado, USA. (October 20–24, 2025)
- **Pokhrel, T.,** Mehan, S., Trooien, T., Gilbertson, J., & Berg, B. (2025). *Assessment of a Nature-Based Solution: Seasonal Riparian Area Management on Ecology and Economy in Skunk Creek, South Dakota*. 2025 South Dakota Student Water Conference, Brookings, South Dakota. (October 15, 2025)
- **Pokhrel, T.,** & Mehan, S. (2025). *Integrated assessment of seasonal riparian area management on vegetation health and water quality*. 2025 ASABE North Central Intersectional Conference. (April 25, 2025)

RESEARCH/PUBLICATIONS EXPERIENCE

MANUSCRIPT

- **Pokhrel, T.,** Mehan, S., Berg, B., & Gilbertson, J. (2025). *Assessment of ecological and economic benefits of seasonal riparian area management in eastern South Dakota.* (Published: <https://doi.org/10.1002/eco.70246>)
- **Pokhrel, T.,** Mehan, S., Lamichhane, M., & Shrestha, E (2025). *Assessment of drought by analyzing the spatiotemporal distribution of ET using remote sensing across heterogeneous climatic regions of Nepal.* (under review)
- **Pokhrel, T.,** Mehan, S., Berg, B., & Gilbertson, J. (2025). *Evaluating spatial gaps in riparian management using SWAT+: Sediment and nitrate transport in the Skunk Creek watershed, South Dakota.* (under review)

HONORS AND AWARDS

- 2026 ASABE Annual International Meeting Travel Grant sponsored by the Agricultural and Biosystems Engineering Department at South Dakota State University, Brookings, SD 2026
- Runners-up at position in the student poster competition at the 2026 SD ASABE Section meeting, Brookings, SD, April 2, 2026 2026
- Honored with gold medal being Department Topper in Agricultural Engineering, IOE, for outstanding academic performance (2016-2021) 2021
- Hutta Ram Baidya Scholarship-Awarded by the Nepalese Society of Agricultural Engineers for securing first position in B.E. Agricultural Engineering 2021
- NAESS Alumni Economic Hardship Scholarship-Awarded by Agricultural Engineers' Alumni in the USA for excellent academic performance 2019
- NAESS Alumni Economic Hardship Scholarship-Awarded by Agricultural Engineers' Alumni in the USA for excellent academic performance 2018

Leadership

Vice President

Agricultural and Biosystems Engineering Graduate Student Association (ABEGSA)

May 2026-Present

- *Elected Vice President to provide leadership, coordinate graduate student events, and promote student engagement and professional development.*

PROFESSIONAL MEMBERSHIPS AND ACTIVITIES

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| Member of the American Geophysical Union (AGU) | 2025-Present |
| • Member of the American Society of Agricultural and Biological Engineers | 2025-Present |
| • Member of Universities Council on Water Resources (UCOWR) and National Institutes for Water Resources (NIWR) | 2025-Present |